Quick tutorial for linear Gaussian state-space analysis of temperature time series with ThermoSST

Akira Hirao

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Set ennvironment

```r
## Set libraries
library(ThermoSSM)

library(tidyverse)
library(forecast)
library(cowplot)

# initialize
rm(list=ls(all=TRUE))
```

Loading test data

```r
# loading monthly mean temperature at the summit of Mt. Fuji
data(fuji_temp)
head(fuji_temp)
```

```
##        Jul   Aug   Sep   Oct   Nov   Dec
## 1932   5.5   5.7   1.8  -4.6  -9.5 -12.9
```

Check of ts object

```r
frequency(fuji_temp)   # frequency 12
```

```
## [1] 12
```

```r
start(fuji_temp)       # start time 1982, 1
```

```
## [1] 1932    7
```

```r
end(fuji_temp)         # end time 2025, 10
```

```
## [1] 2025   12
```

```r
cycle(fuji_temp) %>% head() #     1 12
```

```
##      Jul Aug Sep Oct Nov Dec
## 1932   7   8   9  10  11  12
```

```r
time(fuji_temp)  %>% head() #   1982.000, 1982.083...
```

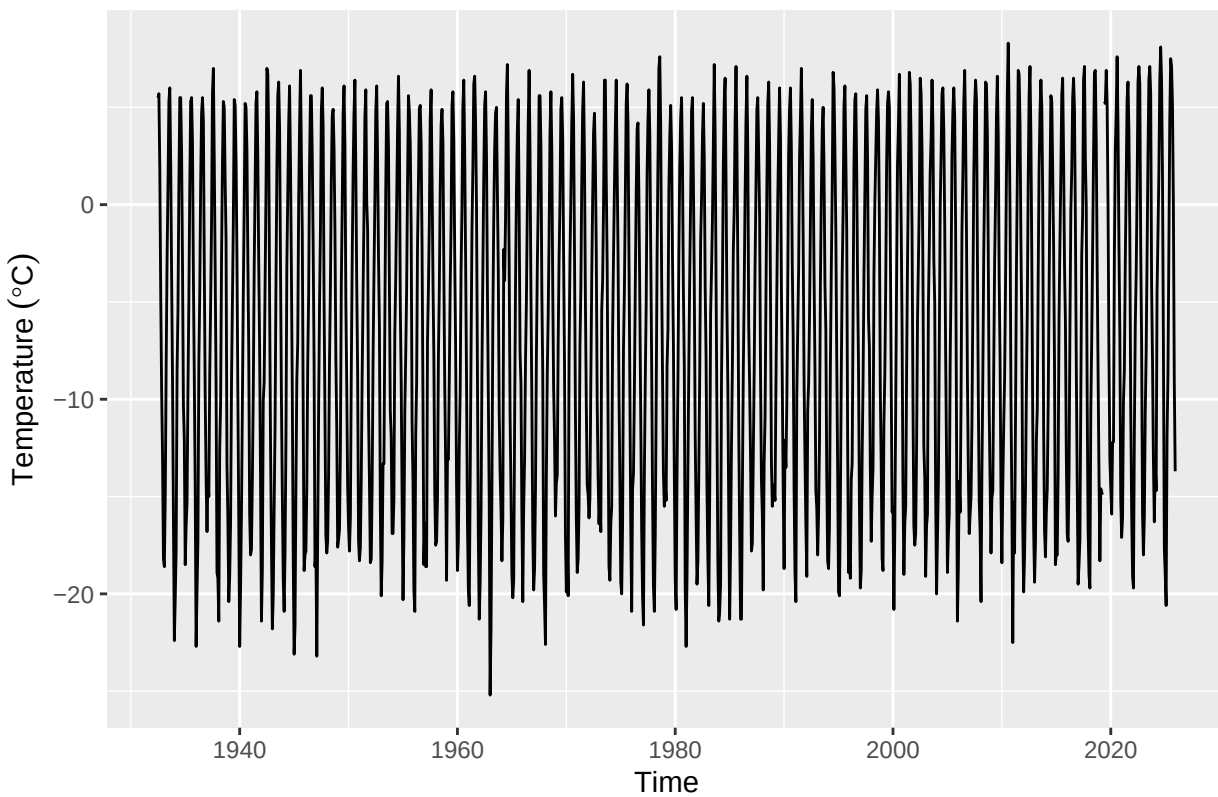
```
##           Jul      Aug      Sep      Oct      Nov      Dec
## 1932 1932.500 1932.583 1932.667 1932.750 1932.833 1932.917
```

```r
window(fuji_temp, start = c(2001, 1), end = c(2001, 12))  #
```

```
##        Jan   Feb   Mar   Apr   May   Jun   Jul   Aug   Sep   Oct   Nov   Dec
## 2001 -19.0 -16.4 -15.5  -8.5  -3.0   1.6   6.8   6.2   3.1  -1.4 -10.0 -16.5
```

Plotting time series of temperature

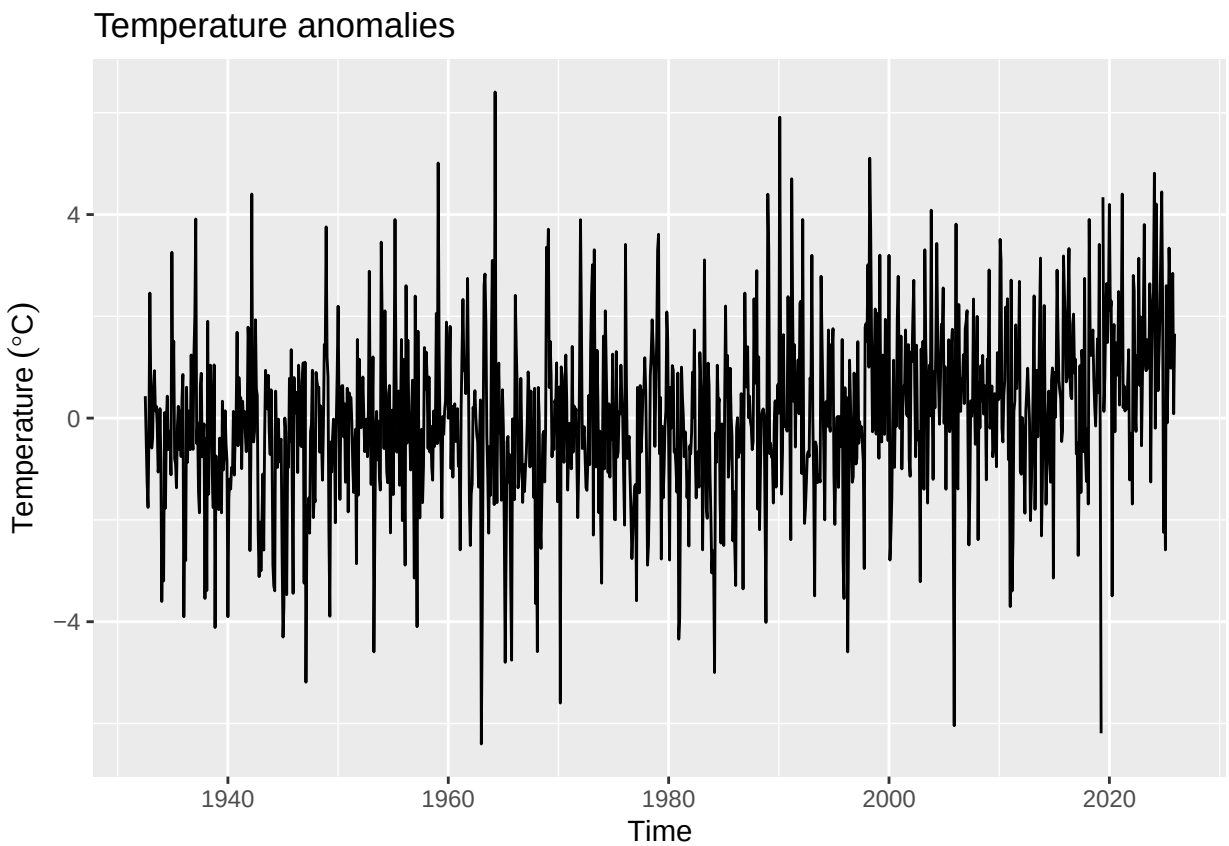
```r
fuji_temp_plot <- forecast::autoplot(fuji_temp) +
  labs(y = expression(Temperature~(degree*C)), x = "Time")

plot(fuji_temp_plot)
```




```r
#ggsave("Fuji_temp_plot.png",
#       width=8,height=4)
```

Plotting time series of temperature deviation

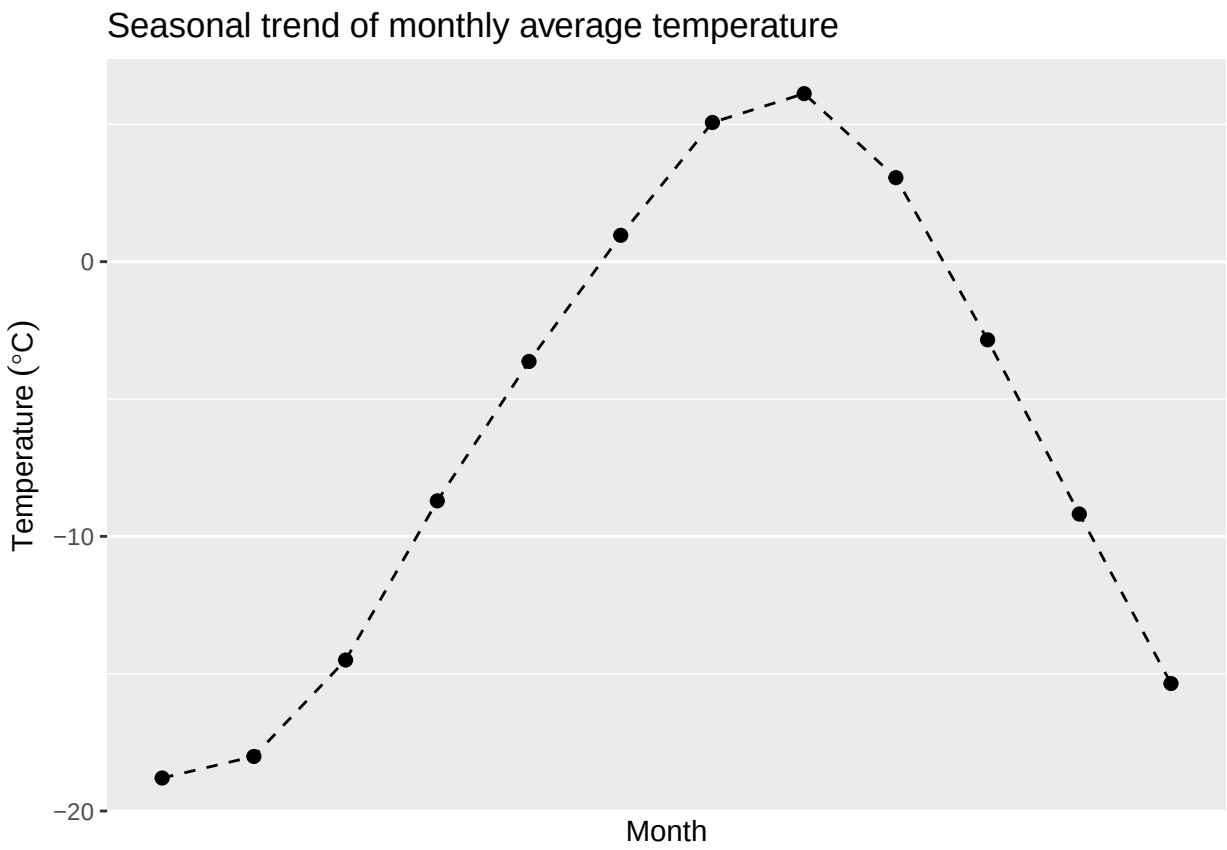
```r
plot_temp_dev_obj <- plot_temp_dev(fuji_temp)
```




```
#ggsave("Fuji_temp_dev_plot.png",
#       width=8,height=4)
```

Plotting seasonal pattern of monthly temperature

```
plot_typical_seasonal_cycle(fuji_temp)
```




```
#ggsave("Fuji_ave_monthly_temp_plot.png",
#       width=8,height=6)
```

線形ガウス状態空間モデルの適用

```
res <- lgssm(fuji_temp)
fit    <- res[[1]] # fit model
smooth <- res[[2]] # smoothing

#   95%
ci <- confint(smooth, level = 0.95)
```

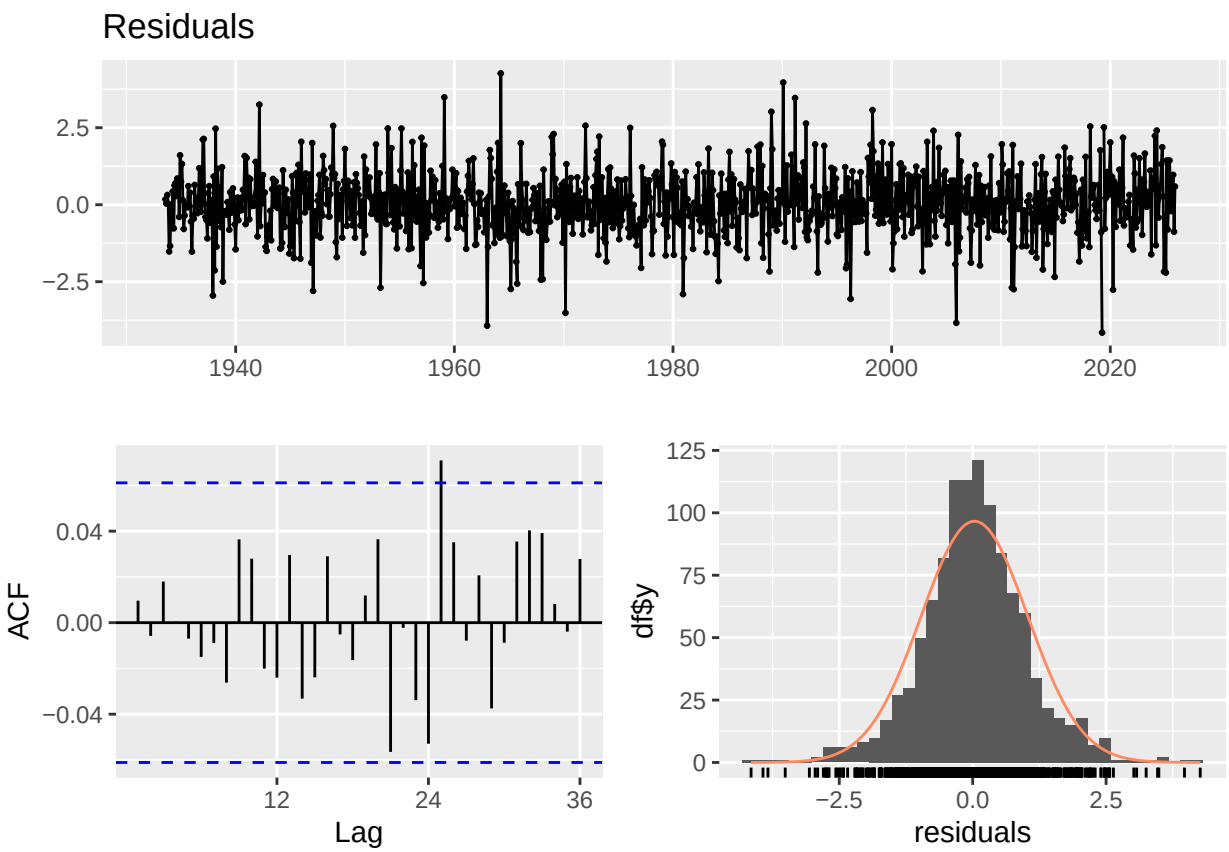
簡単なモデル診断

```
##
# 0
fit$optim.out$convergence
```

```
## [1] 0
```

```
##
#
std_obs_resid <- rstandard(smooth, type = "recursive")

# forecast   checkredisuals
```

```r
# Ljung-Box : P > 0.05
checkresiduals(std_obs_resid)
```




```
##
##  Ljung-Box test
##
## data:  Residuals
## Q* = 17.591, df = 24, p-value = 0.8224
##
## Model df: 0.   Total lags used: 24
```

```r
#



##
# P > 0.05
shapiro.test(std_obs_resid)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  std_obs_resid
## W = 0.98378, p-value = 9.022e-10
```

モデルの推定値

```
#     (      )
pars_comp <- extract_pars(res)
pars_comp
```

```
##       Q_trend      Q_season           AR1           AR2          Q_ar
##  8.951445e-09  5.177007e-04  2.141508e-01 -8.769470e-03  2.443921e+00
##             H
##  1.374109e-07
```

```
#
alpha_hat <- smooth$alphahat
head(alpha_hat)
```

```
##              level        slope sea_dummy1 sea_dummy2 sea_dummy3 sea_dummy4
## Jul 1932 -6.816754 0.0005590682  11.495327   7.219398   2.636248  -2.286643
## Aug 1932 -6.816195 0.0005590711  12.426540  11.495327   7.219398   2.636248
## Sep 1932 -6.815636 0.0005590772   9.344077  12.426540  11.495327   7.219398
## Oct 1932 -6.815077 0.0005590846   3.450236   9.344077  12.426540  11.495327
## Nov 1932 -6.814518 0.0005590890  -2.895071   3.450236   9.344077  12.426540
## Dec 1932 -6.813959 0.0005590901  -8.902299  -2.895071   3.450236   9.344077
##          sea_dummy5 sea_dummy6 sea_dummy7 sea_dummy8 sea_dummy9 sea_dummy10
## Jul 1932  -8.214384 -11.712963 -12.560467  -8.902299  -2.895071    3.450236
## Aug 1932  -2.286643  -8.214384 -11.712963 -12.560467  -8.902299   -2.895071
## Sep 1932   2.636248  -2.286643  -8.214384 -11.712963 -12.560467   -8.902299
## Oct 1932   7.219398   2.636248  -2.286643  -8.214384 -11.712963  -12.560467
## Nov 1932  11.495327   7.219398   2.636248  -2.286643  -8.214384  -11.712963
## Dec 1932  12.426540  11.495327   7.219398   2.636248  -2.286643   -8.214384
##          sea_dummy11     arima1        arima2
## Jul 1932    9.344077  0.8214272 -0.0015357366
## Aug 1932    3.450236  0.0896555 -0.0072034814
## Sep 1932   -2.895071 -0.7284408 -0.0007862312
## Oct 1932   -8.902299 -1.2351588  0.0063880402
## Nov 1932  -12.560467  0.2095890  0.0108316881
## Dec 1932  -11.712963  2.8162577 -0.0018379845
```

```
level <- alpha_hat[,"level"]
drift <- alpha_hat[,"slope"]

level_ts <- ts(level, start = start(fuji_temp), frequency = 12)
drift_ts <- ts(drift, start = start(fuji_temp), frequency = 12)

#
mean_drift_year <- mean(drift_ts) * 12
print(mean_drift_year)
```

```
## [1] 0.01719615
```

```
# 1940
drift_1940s_per_year <- window(drift_ts,
                               start=c(1940,1),
                               end=c(1949,12)
                               ) %>%  mean()*12
print(drift_1940s_per_year)
```

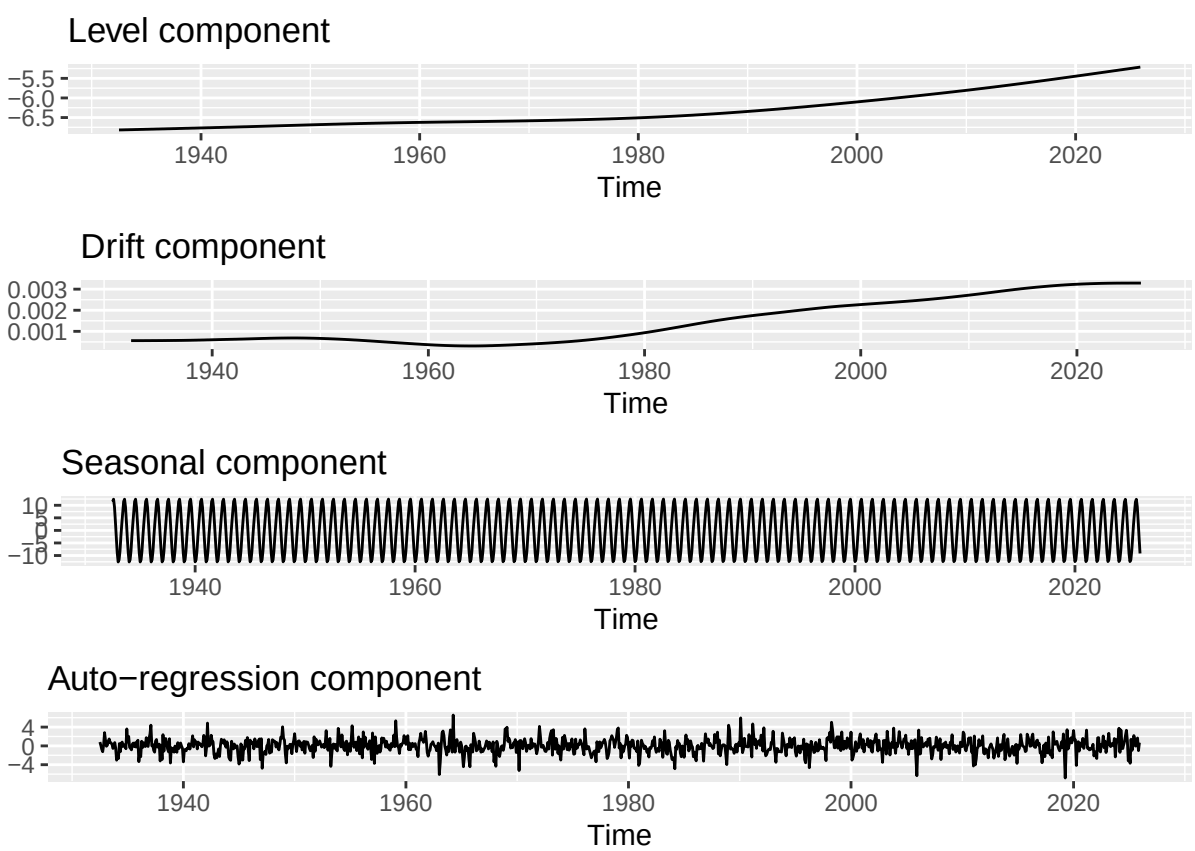
```
## [1] 0.007860923
```

```r
# 2010
drift_2010s_per_year <- window(drift_ts,
                               start=c(2010,1),
                               end=c(2019,12)
                               ) %>%  mean()*12
print(drift_2010s_per_year)
```

```
## [1] 0.03606574
```

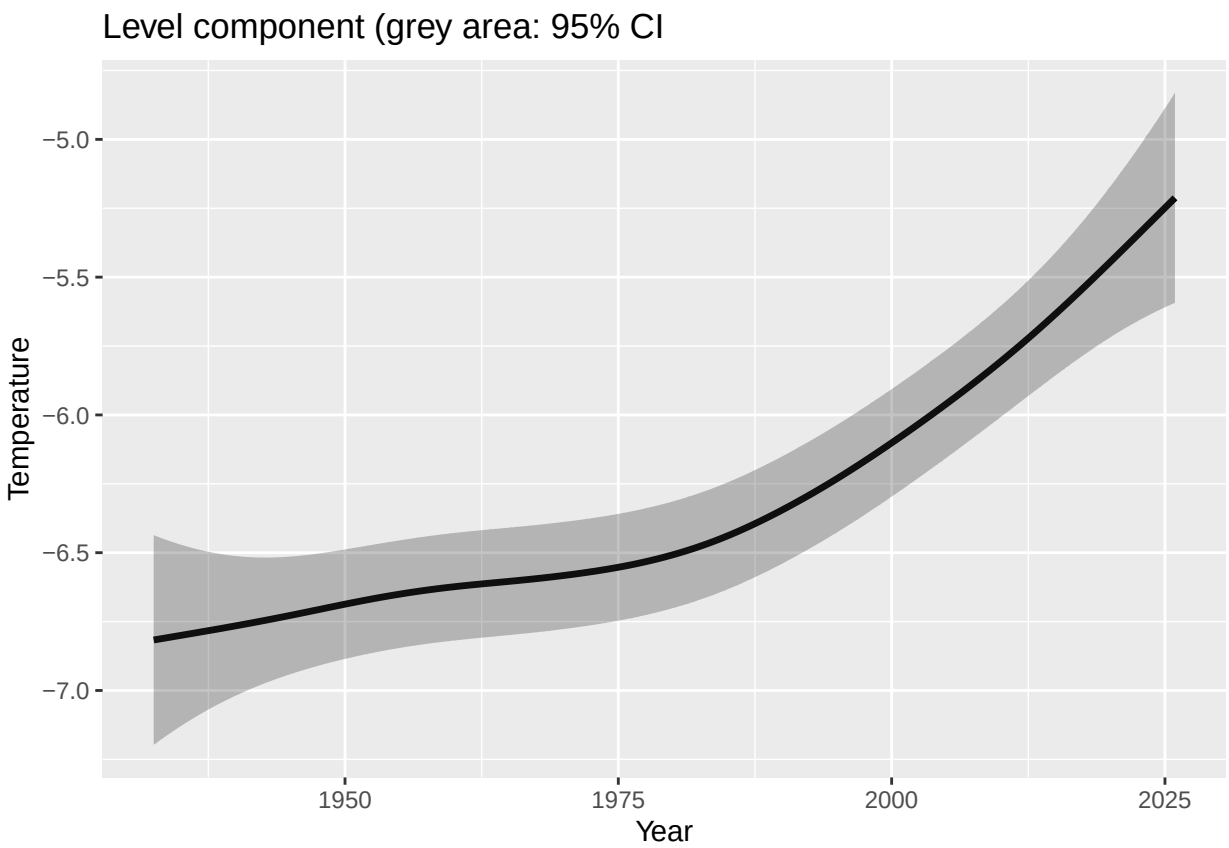
推定結果の各成分（水準、ドリフト、季節周期、定常AR）のプロット

```r
plt_comp <- ThermoSSM::plot_level_trend_season_ar(res)
plot_four_comp <- cowplot::plot_grid(plt_comp[[1]],
                   plt_comp[[2]],
                   plt_comp[[3]],
                   plt_comp[[4]],
                   ncol=1)

ggsave("plot_level_trend_season_ar.png",
       plot=plot_four_comp,
       width=8,height=8)
plot_four_comp
```



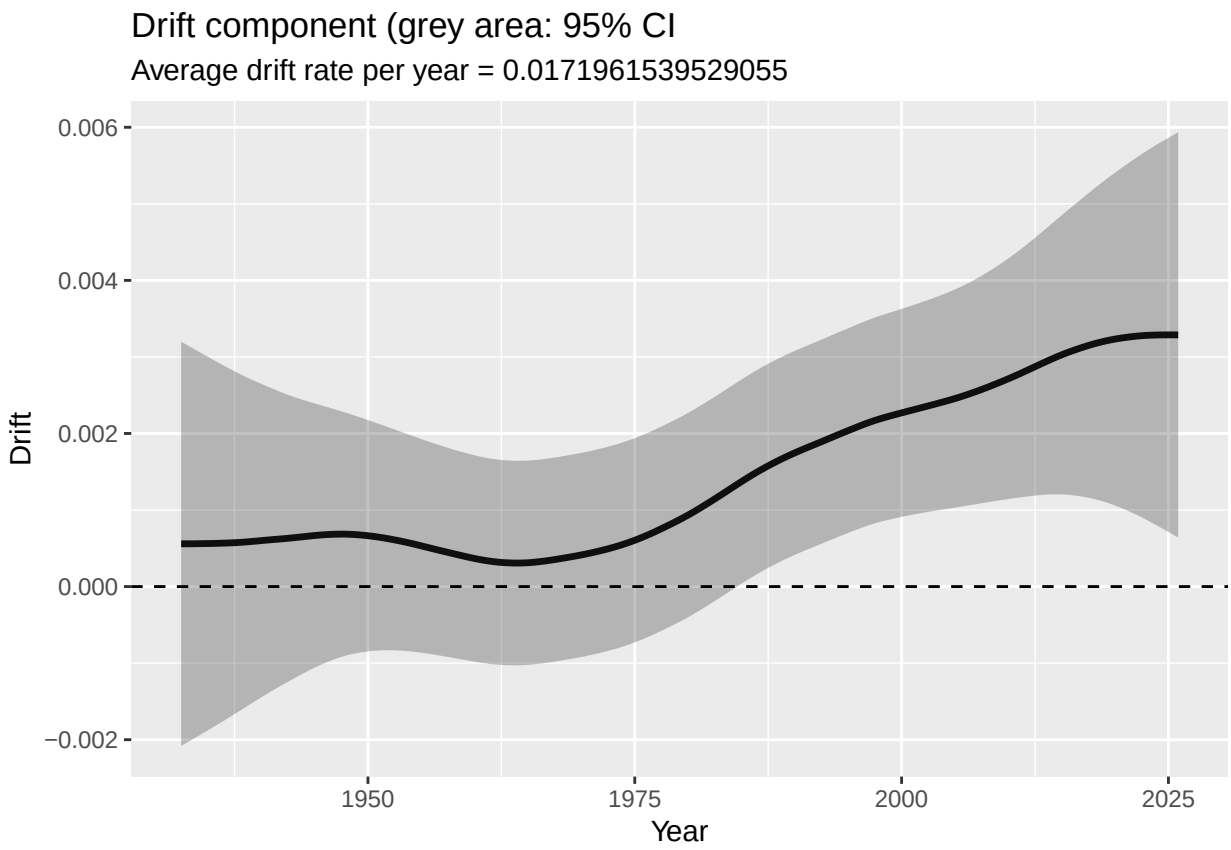
水準変動の信頼区間の図示

```
plt_level_ci <- plot_level_ci(res)

ggsave("level_plot.png",
       width=6, height=4,
       plot = plt_level_ci)

plot(plt_level_ci)
```



ドリフト成分の図示

```
plt_drift_ci <- plot_drift_ci(res)

ggsave("drift_plot.png",
       width=6, height=4,
       plot = plt_drift_ci)

plot(plt_drift_ci)
```


Drift component (grey area: 95% CI
Average drift rate per year = 0.0171961539529055
0.006
0.004
0.002
0.000
−0.002
Drift
1950
1975
2000
2025
Year